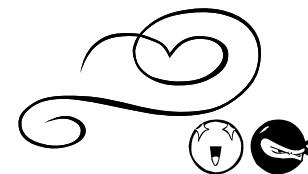




Integration selective



Q 1

Abbotsleigh

Use the substitution $u = \ln x$ to evaluate $\int_e^{e^2} \frac{1}{x \ln x} dx$.

$$\begin{aligned} (a) \quad & \int_e^{e^2} \frac{1}{x \ln x} dx \\ &= \int_1^2 \frac{1}{u} \frac{du}{dx} dx \\ &= \int_1^2 \frac{1}{u} du \\ &= [\ln u]_1^2 \\ &= \ln 2 - \ln 1 \\ &= \ln 2 \end{aligned}$$

$$\begin{aligned} u &= \ln x \\ \frac{du}{dx} &= \frac{1}{x} \\ x=e \quad u &= \ln e \\ &= 1 \\ x=e^2 \quad u &= \ln e^2 \\ &= 2 \ln e \\ &= 2 \end{aligned}$$

Q 2

Abbotsleigh

(i) Find $\int \frac{x}{x+9} dx$

2

(ii) Evaluate $\int_0^4 x \sqrt{x^2+9} dx$ using the substitution $u = x^2+9$

3

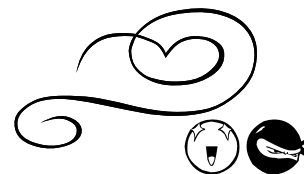
$$\begin{aligned} (c) \quad (i) \quad & \int \frac{x}{x+9} dx = \int \frac{x+9-9}{x+9} dx \\ &= \int \left(\frac{x+9}{x+9} - \frac{9}{x+9} \right) dx \\ &= \int \left(1 - \frac{9}{x+9} \right) dx \\ &= x - 9 \ln(x+9) + c \end{aligned}$$

$$\begin{aligned} (ii) \quad & \int_0^4 x \sqrt{x^2+9} dx \quad \begin{array}{l} u = x^2+9 \\ \frac{du}{dx} = 2x \\ x=0 \quad u=9 \\ x=4 \quad u=25 \end{array} \\ &= \frac{1}{2} \int_9^{25} \sqrt{u} \cdot 2 du \\ &= \frac{1}{2} \int_9^{25} \sqrt{u} du \\ &= \frac{1}{2} \int_9^{25} u^{\frac{1}{2}} du \\ &= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_9^{25} \\ &= \frac{1}{3} \left((\sqrt{25})^3 - (\sqrt{9})^3 \right) \\ &= \frac{1}{3} (125 - 27) \\ &= \frac{98}{3} \end{aligned}$$





Integration selective



Q 3

Abbotsleigh

Find $\int \sin^2 x \cos^2 x \, dx$

$$\begin{aligned} \int \sin^2 x \cos^2 x \, dx \\ &= \int (\sin x \cos x)^2 \, dx \\ &= \int \left(\frac{1}{2} \sin 2x\right)^2 \, dx \end{aligned}$$

$$\begin{aligned} \int \sin^2 x \cos^2 x \, dx &= \frac{1}{4} \int \sin^2 2x \, dx \\ &= \frac{1}{4} \int \frac{1}{2} (1 - \cos 4x) \, dx \\ &= \frac{1}{8} \left(x - \frac{1}{4} \sin 4x\right) + C \\ &= \frac{x}{8} - \frac{1}{32} \sin 4x + C \end{aligned}$$

Q 4

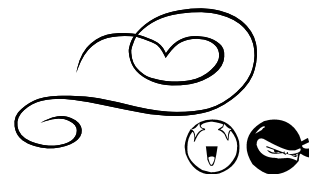
Abbotsleigh

Evaluate $\int_0^1 x(2-x)^3 \, dx$ using the substitution $u = 2-x$

$$\begin{aligned} \text{(e)} \quad \int_0^1 x(2-x)^3 \, dx & \quad \left. \begin{array}{l} u = 2-x \\ du = -dx \end{array} \right\} \\ &= \int_2^1 (2-u)(u)^3 (-du) \quad \left. \begin{array}{l} x=0 \quad u=2 \\ x=1 \quad u=1 \end{array} \right\} \\ &= \int_1^2 2u^3 - u^4 \, du \\ &= \left[\frac{2u^4}{4} - \frac{u^5}{5} \right]_1^2 \\ &= \frac{2 \times 2^4}{4} - \frac{2^5}{5} - \left(\frac{2 \times 1^4}{4} - \frac{1}{5} \right) \\ &= 1\frac{3}{10} \quad \text{or} \quad \underline{\underline{\frac{13}{10}}} \end{aligned}$$



Integration selective



Q 5

Abbotsleigh

Find $\int \sin^2 6x \, dx$

$$\begin{aligned}\int \sin^2 6x \, dx &= \frac{1}{2} \int (1 - \cos 12x) \, dx \\ &= \frac{1}{2} \left[x - \frac{1}{12} \sin 12x \right] \\ &= \frac{x}{2} - \frac{\sin 12x}{24} + c\end{aligned}$$

Q 6

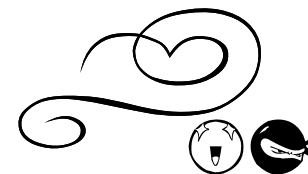
Abbotsleigh

Using the substitution $u = x + 2$, evaluate $\int_{-1}^2 \frac{x}{3} \sqrt{x+2} \, dx$.

$$\begin{aligned}\text{e) } \int_{-1}^2 \frac{x}{3} \sqrt{x+2} \, dx &= I \\ \text{Let } u &= x + 2 \Rightarrow x = u - 2 \\ \frac{du}{dx} &= 1 \Rightarrow du = dx\end{aligned}$$

$$\begin{aligned}x: -1 &\rightarrow 2 \\ u: 1 &\rightarrow 4\end{aligned} \quad \left. \vphantom{\begin{aligned}x: -1 &\rightarrow 2 \\ u: 1 &\rightarrow 4\end{aligned}} \right\} \text{limits}$$

$$\begin{aligned}I &= \int_1^4 \frac{u-2}{3} \sqrt{u} \, du \\ &= \frac{1}{3} \int_1^4 (u^{\frac{3}{2}} - 2u^{\frac{1}{2}}) \, du \\ &= \frac{1}{3} \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} \right]_1^4 \\ &= \frac{1}{3} \left(\left(\frac{2}{5} \times 32 - \frac{4}{3} \times 8 \right) - \left(\frac{2}{5} - \frac{4}{3} \right) \right) \\ &= \frac{1}{3} \left(\frac{62}{5} - \frac{28}{3} \right) \\ &= \frac{46}{45} \quad (\text{or } 1.02)\end{aligned}$$



Q 7

Abbotsleigh

Use the substitution $u = 2 - x^2$ to find $\int \frac{x}{(2-x^2)^3} dx$

$$d) \quad \begin{aligned} u &= 2 - x^2 \\ \frac{du}{dx} &= -2x \end{aligned}$$

$$\therefore x dx = \frac{-du}{2}$$

$$\int \frac{x dx}{(2-x^2)^3} = -\frac{1}{2} \int \frac{du}{u^3}$$

$$= -\frac{1}{2} \int u^{-3} du$$

$$= -\frac{1}{2} \frac{u^{-2}}{-2} + C$$

$$= \frac{1}{4(2-x^2)^2} + C$$

Q 8

Abbotsleigh

Show that $\int_{\pi/4}^{\pi/3} \sin^2 3x dx = \frac{1}{2} \left(\frac{\pi}{12} - \frac{1}{6} \right)$

$$\int_{\pi/4}^{\pi/3} \sin^2 3x dx = \frac{1}{2} \int_{\pi/4}^{\pi/3} 1 - \cos 6x dx$$

$$= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right]_{\pi/4}^{\pi/3}$$

$$\cos 2x = 1 - 2\sin^2 x$$

$$\cos 6x = 1 - 2\sin^2 3x$$

$$2\sin^2 3x = 1 - \cos 6x$$

$$\sin^2 3x = \frac{1}{2} (1 - \cos 6x)$$

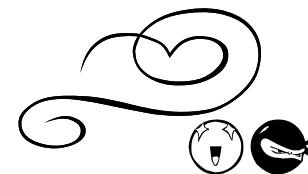
$$= \frac{1}{2} \left[\frac{\pi}{3} - \frac{1}{6} \sin 2\pi \right] - \left(\frac{\pi}{4} - \frac{1}{6} \sin \frac{3\pi}{2} \right)$$

$$= \frac{1}{2} \left[\frac{\pi}{3} - \frac{\pi}{4} - \frac{1}{6} \right]$$

$$= \frac{1}{2} \left[\frac{4\pi - 3\pi}{12} - \frac{1}{6} \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{12} - \frac{1}{6} \right] \text{ as required.}$$





Q 9

Ascham

Using the substitution $u = x^3 + 1$, evaluate $\int_{-1}^1 x^2 (x^3 + 1) dx$

$$\int_{-1}^1 x^2 (x^3 + 1) dx$$

$$\frac{1}{3} \int_0^2 u du$$

$$\frac{1}{3} \left[\frac{1}{2} u^2 \right]_0^2$$

$$= \frac{1}{6} (4 - 0)$$

$$\boxed{\frac{2}{3}}$$

$$\begin{aligned} u &= x^3 + 1 \\ du &= 3x^2 dx \\ \therefore x^2 dx &= \frac{1}{3} du \\ \text{when } x=1, u &= 2 \\ x=-1, u &= 0 \end{aligned}$$

Q10

Ascham

Use the substitution $u = 1 + x$ to find $\int_1^2 \frac{1-x}{(1+x)^3} dx$

$$\int_1^2 \frac{1-x}{(1+x)^3} dx$$

$$= \int_2^3 \frac{1-(u-1)}{u^3} du$$

$$= \int_2^3 \frac{2-u}{u^3} du \quad \checkmark$$

$$= \int_2^3 \frac{2}{u^3} - \frac{1}{u^2} du$$

$$= \left[\frac{2u^{-2}}{-2} - \frac{u^{-1}}{-1} \right]_2^3$$

$$= \left[-\frac{1}{u^2} + \frac{1}{u} \right]_2^3 \quad \checkmark$$

$$= -\frac{1}{9} + \frac{1}{3} - \left(-\frac{1}{4} + \frac{1}{2} \right)$$

$$= -\frac{1}{36} \quad \checkmark$$

$$\begin{aligned} \text{Let } u &= 1+x \\ du &= dx \end{aligned}$$

$$\begin{aligned} x=1, u &= 2 \\ x=2, u &= 3 \\ x &= u-1 \end{aligned}$$

(3)



Q11

Ascham

Find $\int \sin^2 2x dx$

a) $\int \sin^2 2x dx$

$$\cos 4x = 1 - 2\sin^2 2x$$

$$2\sin^2 2x = 1 - \cos 4x$$

$$= \frac{1}{2} \int (1 - \cos 4x) dx$$

$$= \frac{1}{2} \left[x - \frac{1}{4} \sin 4x \right] + C$$

$$= \frac{1}{2} x - \frac{1}{8} \sin 4x + C$$

(2)

Q12

Ascham

Evaluate $\int_{\frac{1}{2}}^1 4t(2t-1)^3 dt$ by using the substitution $u = 2t - 1$.

d) $\int_{\frac{1}{2}}^1 4t(2t-1)^3 dt$

$$= \int_0^1 2(u+1)(u)^3 \cdot \frac{du}{2}$$

$$= \int_0^1 u^4 + u^3 du$$

$$= \left[\frac{u^5}{5} + \frac{u^4}{4} \right]_0^1$$

$$= \frac{1}{5} + \frac{1}{4} - 0$$

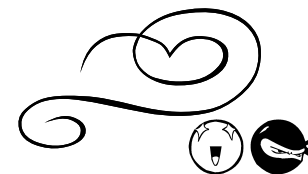
$$= \frac{9}{20}$$

u = 2t - 1
du = 2dt
t = 1 u = 1
t = 1/2 u = 0

(4)



Integration selective



Q13

Barker

Use the substitution $u^2 = x + 4$ (where $u > 0$) to evaluate $\int_{-3}^0 \frac{x}{\sqrt{x+4}} dx$

$$b) u^2 = x + 4$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$x = -3, u = 1$$

$$x = 0, u = 2$$

$$\therefore I = \int_1^2 \frac{u^2 - 4}{\sqrt{u^2}} \cdot 2u du$$

$$= \int_1^2 \frac{u^2 - 4}{u} \times 2u du$$

$$= 2 \left[\frac{u^3}{3} - 4u \right]_1^2$$

$$= 2 \left[\left(\frac{8}{3} - 8 \right) - \left(\frac{1}{3} - 4 \right) \right]$$

$$I = \frac{-10}{3} \checkmark$$

Q14

Barker

Using the substitution $u = x^4 + 8$, or otherwise, find $\int \frac{4x^3 dx}{\sqrt{x^4 + 8}}$

$$b) u = x^4 + 8$$

$$\frac{du}{dx} = 4x^3$$

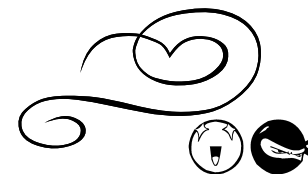
$$\therefore du = 4x^3 dx$$

$$\int \frac{4x^3 dx}{\sqrt{x^4 + 8}} = \int \frac{du}{\sqrt{u}}$$

$$= \int u^{-1/2} du$$

$$= \frac{u^{1/2}}{1/2} + C$$

$$= 2\sqrt{x^4 + 8} + C$$



Q15

Barker

Find $\int_{-3}^0 x\sqrt{1-x} dx$ using the substitution $u = 1-x$

$$\begin{aligned}
 (d) \quad u &= 1-x \rightarrow x = 1-u \\
 du &= -dx \\
 x = -3, u &= 4 \\
 x = 0, u &= 1 \\
 \therefore \int_{-3}^0 (1-u)\sqrt{u} du &= \left(\frac{2}{3}u^{3/2} - \frac{2}{5}u^{5/2} \right) \Big|_1^4 \\
 &= \frac{2}{3}(8) - \frac{2}{5}(32) - \frac{2}{3} + \frac{2}{5} \\
 &= -7\frac{11}{15} \\
 \therefore \int_{-3}^0 (1-u)\sqrt{u} du &= \int_1^4 \sqrt{u} - u^{3/2} du \\
 &= \int_1^4 \sqrt{u} - u^{3/2} du
 \end{aligned}$$

Q16

Barker

(i) Using $\cos(2ax) = 1 - 2\sin^2(ax)$, find the simplest expression for

$$\int_0^{\pi/2} \sin^2(ax) dx$$

(ii) If a is an integer, $\int_0^{\pi/2} \sin^2(ax) dx$ always gives the same number.

Find this number.

$$\begin{aligned}
 (i) \quad I &= \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2ax) dx \\
 &= \frac{1}{2} \left[x - \frac{\sin 2ax}{2a} \right]_0^{\pi/2} \\
 &= \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{\sin a\pi}{2a} \right) - 0 \right] \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - \frac{\sin a\pi}{2a} \right]
 \end{aligned}$$

(ii) Since $\sin a\pi = 0$

$$\text{then } \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$$

Q17

Barker

Find $\int 6x^3 \sqrt{(3x^4-3)^3} dx$ using the substitution $u = 3x^4-3$

(a) $u = 3x^4 - 3$

$\therefore \frac{du}{dx} = 12x^3 \quad \therefore \frac{du}{2} = 6x^3 dx$

$\therefore \int 6x^3 \sqrt{(3x^4-3)^3} dx = \int \frac{1}{2} \sqrt{u^3} du$

$= \frac{1}{2} \int u^{3/2} du$

$= \frac{1}{2} \times \frac{2}{5} u^{5/2} + C$

$= \frac{1}{5} (3x^4-3)^{5/2} + C \quad \text{or} \quad \frac{\sqrt{(3x^4-3)^5}}{5} + C$

Q18

Barker

Use the substitution $x = 2 \sin \theta$ to evaluate $\int_0^1 \sqrt{4-x^2} dx$

b) $x = 2 \sin \theta$

$dx = 2 \cos \theta d\theta$

when $x=0$, $\theta=0$

$x=1$, $\sin \theta = \frac{1}{2}$, $\theta = \frac{\pi}{6}$

$\therefore \int_0^{\pi/6} \sqrt{4-(2\sin\theta)^2} \times 2 \cos \theta d\theta$

$= \int_0^{\pi/6} \sqrt{4(1-\sin^2\theta)} \times 2 \cos \theta d\theta$

$= \int_0^{\pi/6} \sqrt{4 \cos^2 \theta} \times 2 \cos \theta d\theta$

$= \int_0^{\pi/6} 2 \cos \theta \times 2 \cos \theta d\theta$

$= 4 \int_0^{\pi/6} \cos^2 \theta d\theta$

now $\cos^2 \theta = \frac{1+\cos 2\theta}{2}$

$\therefore = 4 \int_0^{\pi/6} \frac{1+\cos 2\theta}{2} d\theta$

$= 2 \int_0^{\pi/6} 1+\cos 2\theta d\theta$

$= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/6}$

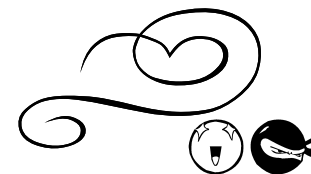
$= 2 \left(\frac{\pi}{6} + \frac{\sin \frac{\pi}{3}}{2} - 0 - 0 \right)$

$= 2 \left(\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right)$

$= \frac{\pi}{3} + \frac{\sqrt{3}}{2}$



Integration selective



Q19

Barker

Evaluate $\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx$

$$a) \left[\frac{\sin^3 x}{3} \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{3} \left(\sin^3 \frac{\pi}{4} - \sin^3 0 \right)$$

$$= \frac{1}{3} \times \left(\frac{1}{\sqrt{2}} \right)^3$$

$$= \frac{1}{3} \times \frac{1}{2\sqrt{2}}$$

$$= \frac{1}{6\sqrt{2}} \quad \left(\text{or } \frac{\sqrt{2}}{12} \right)$$

Q20

Barker

By using the substitution $u = \sqrt{x}$, evaluate

$$\int_1^4 \frac{dx}{x + \sqrt{x}}$$

$$a. \int_1^4 \frac{dx}{x + \sqrt{x}}$$

$$u = x^{\frac{1}{2}} \quad \text{at } x=4 \quad u=2$$

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}} \quad x=1 \quad u=1$$

$$2u \, du = dx$$

$$= \int_1^2 \frac{2u \, du}{u^2 + u}$$

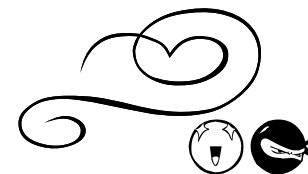
$$= \int_1^2 \frac{2 \, du}{u+1} \quad \checkmark$$

$$= \left[2 \ln(u+1) \right]_1^2$$

$$= 2 \ln 3 - 2 \ln 2$$

$$= 2 \ln \frac{3}{2} \quad \checkmark \quad \text{L.B.}$$

Suggested so
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for Trial &
Ext 1 201
• variations do



Q21

Baulkham

Use the substitution $u = 2 + x$ to evaluate $\int_{-2}^2 x\sqrt{2+x} dx$.

$$\begin{aligned}
 e) \int_{-2}^2 x\sqrt{2+x} dx & \quad u = 2+x \\
 & \quad \frac{du}{dx} = 1 \\
 & \quad du = dx \\
 & \quad \text{① CHANGE LIMITS} \\
 & \quad x = -2 \quad u = 0 \\
 & \quad x = 2 \quad u = 4 \\
 & = \int_0^4 (u-2)\sqrt{u} du \\
 & = \int_0^4 u^{\frac{3}{2}} - 2u^{\frac{1}{2}} du \\
 & = \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} \right]_0^4 \quad \text{①} \\
 & = \left(\frac{2}{5} \cdot 32 - \frac{4}{3} \cdot 8 \right) - (0) \\
 & = 2^{2/5} \quad \text{①}
 \end{aligned}$$

37

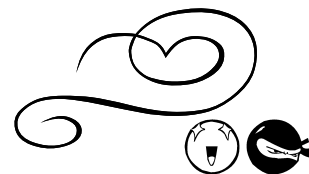
Q22

Baulkham

Use the substitution $u = 2x + 1$ to find $\int x(2x+1)^{10} dx$

$$\begin{aligned}
 e) \int \frac{u-1}{2} \cdot u^{10} \cdot \frac{du}{2} & \quad \checkmark \quad u = 2x+1 \\
 & \quad du = 2dx \\
 & \quad x = \frac{u-1}{2} \\
 & = \frac{1}{4} \int u^{11} - u^{10} du \\
 & = \frac{1}{4} \left[\frac{u^{12}}{12} - \frac{u^{11}}{11} \right] + C \quad \checkmark \\
 & = \frac{1}{48} (2x+1)^{12} - \frac{1}{44} (2x+1)^{11} + C
 \end{aligned}$$





Q23

Baulkham

Find the volume when the area between $y = 2 \sin x$, the x and y axes and $x = \frac{\pi}{4}$ is rotated about the x axis.

$$\begin{aligned}
 V &= \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x \, dx \quad (1) \\
 &= \pi \int_0^{\frac{\pi}{4}} 4 \left(\frac{1}{2} - \frac{\cos 2x}{2} \right) dx \quad (1) \\
 &= 2\pi \int_0^{\frac{\pi}{4}} (1 - \cos 2x) dx \\
 &= 2\pi \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}} \quad (1) \\
 &= 2\pi \left[\frac{\pi}{4} - \frac{\sin \frac{\pi}{2}}{2} \right] - \left[0 - \frac{\sin 0}{2} \right] \\
 &= 2\pi \left[\frac{\pi}{4} - \frac{1}{2} \right] \\
 &= \frac{\pi^2}{2} - \pi \quad (1)
 \end{aligned}$$

Q24

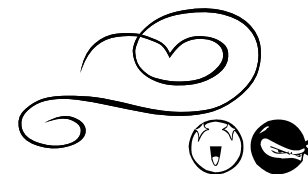
Baulkham

Find $\int \frac{4x}{2x+1} \, dx$ using the substitution $u = 2x + 1$.

$$\begin{aligned}
 b) \int \frac{4x}{2x+1} \, dx & \text{ using } u = 2x + 1 \\
 \frac{du}{dx} &= 2 & x &= \frac{u-1}{2} \\
 dx &= \frac{du}{2} \quad (1) \\
 \int \frac{4 \left(\frac{u-1}{2} \right) \cdot \frac{du}{2}}{u} \\
 &= \int \frac{u-1}{u} \, du \quad (1) \\
 &= \int 1 - \frac{1}{u} \\
 &= u - \ln u + c \quad (1) \\
 &= (2x+1) - \ln(2x+1) + c \\
 & \quad \text{(ignore c)}
 \end{aligned}$$



Integration selective



Q25

Baulkham

$\int_{-1}^0 x\sqrt{1+x} \, dx$, using the substitution $u = 1+x$

$$ii) \int_{-1}^0 x\sqrt{1+x} \, dx$$

$$\left[\begin{array}{l} u=1+x \quad x=u-1 \\ du=dx \\ \text{If } x=-1, u=0 \\ \text{If } x=0, u=1 \end{array} \right] (1)$$

$$= \int_0^1 (u-1) \cdot u^{1/2} \, du \quad (1)$$

$$= \int_0^1 u^{3/2} - u^{1/2} \, du$$

$$= \left[\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_0^1 \quad (1)$$

$$= \left(\frac{2}{5} - \frac{2}{3} \right) - (0 - 0)$$

$$= -\frac{4}{15} \quad (1)$$

Q26

Baulkham

Find $\int \cos^2 2x \, dx$

$$a) \int \cos^2 2x \, dx$$

$$= \int \frac{1}{2} (1 + \cos 4x) \, dx \quad (1)$$

$$= \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c \quad (2)$$

Q27

Baulkham

By using the substitution $w = t^2 - 2$, evaluate $\int_{-1}^{14} \frac{w dw}{\sqrt{w+2}}$

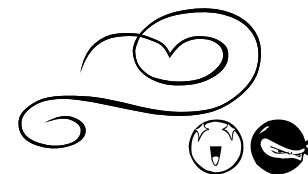
$$\begin{aligned}
 & d) \int_{-1}^{14} \frac{w dw}{\sqrt{w+2}} \\
 & w = t^2 - 2 \quad \frac{dw}{dt} = 2t \\
 & w = 14 \quad t^2 = 16 \quad dw = 2t dt \\
 & \quad \quad \quad t = 4 \\
 & w = -1 \quad t^2 = 1 \\
 & \text{limits } t = 1 \quad \text{①} \\
 & \therefore \int_1^4 \frac{(t^2 - 2) \times 2t dt}{\sqrt{t^2}} \\
 & = 2 \int_1^4 (t^2 - 2) dt \\
 & = 2 \left[\frac{t^3}{3} - 2t \right]_1^4 \\
 & = 2 \left(\left(\frac{64}{3} - 8 \right) - \left(\frac{1}{3} - 2 \right) \right) = \underline{30} \quad \text{①}
 \end{aligned}$$

Q28

Baulkham

Evaluate $\int_0^1 \frac{x dx}{\sqrt{1+x}}$ using the substitution $u^2 = 1+x$.

$$\begin{aligned}
 & d) \int_0^1 \frac{x dx}{\sqrt{1+x}} \\
 & u^2 = 1+x \\
 & x = u^2 - 1 \\
 & \frac{dx}{du} = 2u \\
 & \text{when } x=1, u=\sqrt{2} \quad \checkmark \\
 & \quad \quad \quad x=0, u=1 \\
 & \int_1^{\sqrt{2}} \frac{(u^2 - 1) 2u du}{\sqrt{u^2}} \quad \checkmark \\
 & = 2 \left[\frac{u^3}{3} - u \right]_1^{\sqrt{2}} \\
 & = 2 \left(\left(\frac{2\sqrt{2}}{3} - \sqrt{2} \right) - \left(\frac{1}{3} - 1 \right) \right)
 \end{aligned}$$



Q29

Blackwattle

Use the substitution $u = x^2 + 1$ to evaluate $\int_1^7 \frac{x}{(1+x^2)^2} dx$.

$$u = x^2 + 1$$

$$du = 2x dx$$

$$x = 1 \Rightarrow u = 2$$

$$x = 7 \Rightarrow u = 50$$

$$\begin{aligned} \int_1^7 \frac{x}{(1+x^2)^2} dx &= \frac{1}{2} \int_2^{50} \frac{1}{u^2} du \\ &= \frac{-1}{2} \left[\frac{1}{u} \right]_2^{50} \\ &= \frac{-1}{2} \left(\frac{1}{50} - \frac{1}{2} \right) \\ &= \frac{6}{25} \end{aligned}$$

Q30

Caringbah

Using the substitution $u = x^2 + 1$, or otherwise, find the

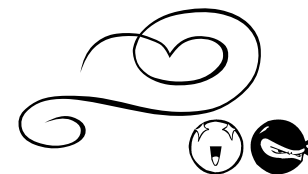
exact value of $\int_0^3 \frac{2x}{x^2+1} dx$.

$$\textcircled{f} \quad \begin{aligned} x=0 &: u=1 \\ x=3 &: u=10 \end{aligned}$$

$$\begin{aligned} \frac{du}{dx} &= 2x \\ \therefore du &= 2x dx \\ \therefore \int_0^3 \frac{2x dx}{x^2+1} &= \int_1^{10} \frac{du}{u} \\ &= [\ln u]_1^{10} \\ &= \ln 10 - \ln 1 \\ &= \ln 10 \end{aligned}$$



Integration selective volumes



Q 1

Abbotsleigh

Find the volume of the solid of revolution formed by rotating the curve $y = x + \frac{1}{x}$ about the x axis between $x = 1$ and $x = 3$.

$$\begin{aligned} V &= \pi \int_1^3 y^2 dx \\ &= \pi \int_1^3 \left(x + \frac{1}{x}\right)^2 dx \\ &= \pi \int_1^3 x^2 + 2 + \frac{1}{x^2} dx \\ &= \pi \int_1^3 x^2 + 2 + x^{-2} dx \\ &= \pi \left[\frac{x^3}{3} + 2x - x^{-1} \right]_1^3 \\ &= \pi \left[\left(\frac{27}{3} + 6 - \frac{1}{3}\right) - \left(\frac{1}{3} + 2 - 1\right) \right] \\ &= \pi \left(\frac{44}{3} - \frac{4}{3} \right) \\ &= \underline{\underline{\frac{40\pi}{3} \text{ cubic units}}} \end{aligned}$$

Q 2

ACE

The area bounded by $y^2 = 3 - 2x - x^2$, $y \geq 0$ and between $x = -3$ and $x = 1$ is revolved about the x axis. Calculate the volume of the solid formed if this area is rotated about the x axis.

$$\begin{aligned} y^2 &= 3 - 2x - x^2 \\ V &= \pi \int_a^b y^2 dx \\ V &= \pi \int_{-3}^1 3 - 2x - x^2 dx \\ &= \pi \left[3x - x^2 - \frac{x^3}{3} \right]_{-3}^1 \\ &= \pi \left[\left(3 - 1 - \frac{1}{3}\right) - (-9 - 9 + 9) \right] \\ &= \frac{32\pi}{3} \end{aligned}$$

Q 3

ACE

The semi-circle $y = \sqrt{4 - x^2}$ is rotated about the x -axis. Which of the following expressions is correct for the volume of the solid of revolution?

- (A) $V = \pi \int_0^2 (4 - x^2) dx$ (C) $V = \pi \int_0^2 (4 - y^2) dy$
(B) $V = 2\pi \int_0^2 (4 - x^2) dx$ (D) $V = 2\pi \int_0^2 (4 - y^2) dy$

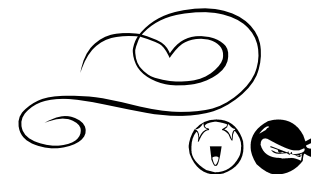
Now $y = \sqrt{4 - x^2}$ or $y^2 = 4 - x^2$

$$\begin{aligned} V &= \pi \int_{-2}^2 y^2 dx \\ &= 2\pi \int_0^2 (4 - x^2) dx \end{aligned}$$

1 Mark: B



Integration selective volumes



Q 4

Abbotsleigh

The semi-circle $y = \sqrt{9 - x^2}$ is rotated about the x -axis. Find the volume of the sphere so formed, in terms of π .

$$V = \pi \int y^2 dx$$

$$V = 2\pi \int_0^3 (\sqrt{9 - x^2})^2 dx$$

$$= 2\pi \left[9x - \frac{x^3}{3} \right]_0^3$$

$$= 2\pi [(27 - 9) - 0]$$

$$= 36\pi \text{ units}^3$$

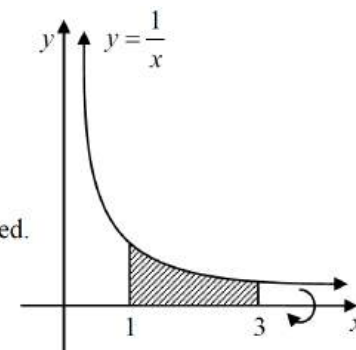
Q 5

ACE

In the diagram, the shaded region is bounded by the curve $y = \frac{1}{x}$, the lines $x = 1$ and $x = 3$, and the x -axis.

The shaded region is rotated about the x -axis.

Calculate the exact volume of the solid of revolution formed.



$$\begin{aligned} V &= \pi \int_1^3 y^2 dx \\ &= \pi \int_1^3 \frac{1}{x^2} dx \\ &= \pi \int_1^3 x^{-2} dx \\ &= -\pi [x^{-1}]_1^3 \\ &= -\pi \left(\frac{1}{3} - 1 \right) \\ &= \frac{2\pi}{3} \text{ units}^3 \end{aligned} \quad (3)$$

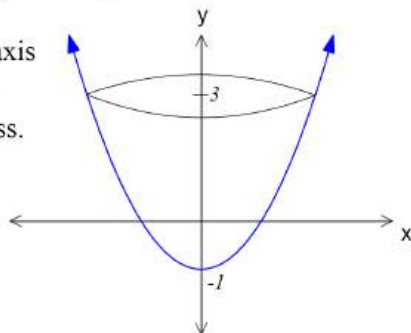
Q 6

Abbotsleigh

The shape of a glass is formed by rotating the curve

$y = 2x^2 - 1$ about the y axis
from $y = -1$ to $y = 3$.

Find the volume of the glass.



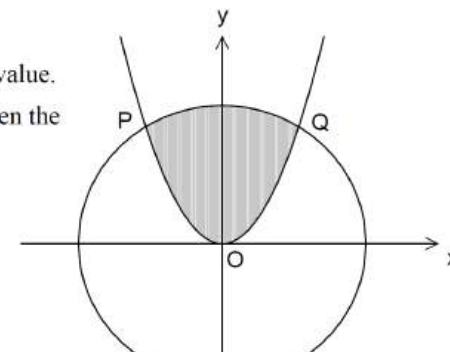
$$\begin{aligned} y &= 2x^2 - 1 \\ x^2 &= \frac{y}{2} + \frac{1}{2} \\ V &= \pi \int_{-1}^3 \left(\frac{y}{2} + \frac{1}{2} \right) dy \quad (1) \\ &= \frac{\pi}{2} \left[\frac{y^2}{2} + y \right]_{-1}^3 \quad (1) \\ &= \frac{\pi}{2} \left[\left(\frac{9}{2} + 3 \right) - \left(\frac{1}{2} - 1 \right) \right] \\ &= 4\pi \text{ units}^3 \end{aligned}$$

Q 7

ACE

The region bounded by the parabola $y = x^2$ and the circle $x^2 + y^2 = 12$ is shown. The parabola and circle intersect at P and Q.

- P and Q have the same y -coordinate. Find its value.
- Find the exact value of the solid generated when the shaded region is rotated about the y -axis.



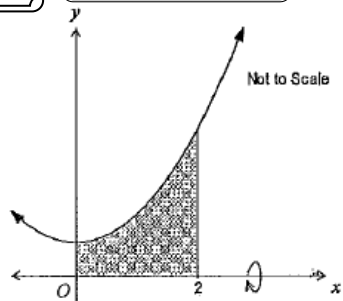
$$\begin{aligned} b) \quad (i) \quad y &= x^2, \quad x^2 + y^2 = 12 \\ y + y^2 &= 12 \\ y^2 + y - 12 &= 0 \\ (y + 4)(y - 3) &= 0 \\ y &= -4 \quad \text{or} \quad y = 3 \end{aligned}$$

$$\begin{aligned} (ii) \quad V_1 &= \pi \int_0^3 y \cdot dy \\ &= \pi \left[\frac{y^2}{2} \right]_0^3 \\ &= \frac{9\pi}{2} \\ V_2 &= \pi \int_3^{4\sqrt{3}} (12 - y^2) \cdot dy \\ &= \pi \left[12y - \frac{y^3}{3} \right]_3^{4\sqrt{3}} \\ &= \pi \left((12 \cdot 4\sqrt{3} - \frac{(4\sqrt{3})^3}{3}) - (36 - 9) \right) \\ &= \pi (16\sqrt{3} - 27) \end{aligned}$$

$$\begin{aligned} \therefore \text{Total vol} &= V_1 + V_2 \\ &= \frac{9\pi}{2} + \pi (16\sqrt{3} - 27) \\ &= \pi \left(\frac{9}{2} + 16\sqrt{3} - 27 \right) \\ &= \pi (16\sqrt{3} - \frac{45}{2}) \end{aligned}$$

Q 8

BAULKHAM



The shaded region in the diagram is bounded by the curve $y = x^2 + 2$, the x -axis, and the lines $x = 0$ and $x = 2$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.

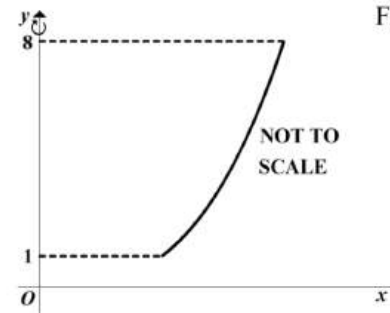
$$\begin{aligned}
 V &= \pi \int_0^2 (x^2 + 2)^2 dx \quad \checkmark \\
 V &= \pi \int_0^2 (x^4 + 4x^2 + 4) dx \\
 &= \pi \left[\frac{x^5}{5} + \frac{4x^3}{3} + 4x \right]_0^2 \quad \checkmark \\
 &= \pi \left(\left[\frac{2^5}{5} + \frac{4 \cdot 2^3}{3} + 4 \cdot 2 \right] - [0] \right) \\
 &= \frac{376\pi}{15} = 78.75 \quad \checkmark
 \end{aligned}$$

Q 9

CALLAGHAN

A solid is formed by rotating the curve $y = x^3$ about the y -axis for $1 \leq y \leq 8$.

Find the volume of the solid in exact form.



$$V = \pi \int_1^8 x^2 dy \quad \left[y = x^3 \rightarrow x^2 = y^{\frac{2}{3}} \right]$$

$$\therefore V = \pi \int_1^8 y^{\frac{2}{3}} dy$$

$$= \pi \left[\frac{3}{5} y^{\frac{5}{3}} \right]_1^8$$

$$= \frac{3\pi}{5} \left(8^{\frac{5}{3}} - 1^{\frac{5}{3}} \right)$$

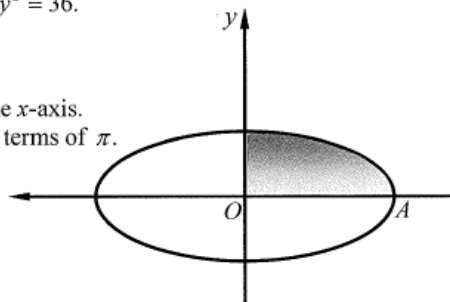
$$= \frac{3\pi}{5} (31) = \frac{93\pi}{5} u^3$$

Q10

CSSA

The diagram below shows an ellipse with equation $4x^2 + 9y^2 = 36$.

- Show that the coordinates of A are $(3,0)$.
- The area shaded in the diagram is rotated about the x -axis. Find the volume generated. Leave your answer in terms of π .



Question 15 (15 marks)

(a)(i) (1 mark)

Sample answer:

$$4x^2 + 9y^2 = 36$$

$$\text{Let } y = 0 \quad 4x^2 = 36 \quad x = 3 \quad \therefore A(3,0)$$

(a)(ii) (3 marks)

Sample answer:

$$\begin{aligned} V &= \pi \int_a^b y^2 dx & V &= \pi \int_0^3 4 - \frac{4}{9}x^2 dx \\ &= \pi \left[4x - \frac{4}{27}x^3 \right]_0^3 = \pi[(12 - 4) - (0)] = 8\pi \text{ units}^3 \end{aligned}$$

Q11

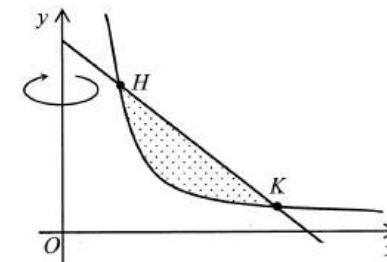
CSSA

The diagram shows the graphs of $y = \frac{2}{x}$ and $y = 3 - x$ for $x > 0$, the shaded area enclosed between the two graphs and their points of intersection H and K as shown.

- Find the coordinates of the points H and K .
- The shaded area is rotated about the y -axis. Show that the volume, V , of the solid formed is given by

$$V = \pi \int_1^2 \left(9 - 6y + y^2 - \frac{4}{y^2} \right) dy$$

- Hence, or otherwise, find the volume V .



(b)(i) (2 marks)

Sample answer

Solve $y = \frac{2}{x}$ and $y = 3 - x$ simultaneously by substitution:

$$\frac{2}{x} = 3 - x$$

$$2 = 3x - x^2$$

$$x^2 - 3x + 2 = 0$$

$$(x - 2)(x - 1) = 0$$

$$x = 2, x = 1$$

$x = 2, y = 1$ and $x = 1, y = 2$

So H is $(1, 2)$ and K is $(2, 1)$.

(b)(ii) (2 marks)

Sample answer

Rotating about the y -axis, so make x the subject of both functions:

$$x = 3 - y \text{ and } x = \frac{2}{y}$$

$$V = \pi \int_1^2 \left((3 - y)^2 - \left(\frac{2}{y} \right)^2 \right) dy$$

$$= \pi \int_1^2 \left(9 - 6y + y^2 - \frac{4}{y^2} \right) dy, \text{ as required}$$

(b)(iii) (2 marks)

Sample answer

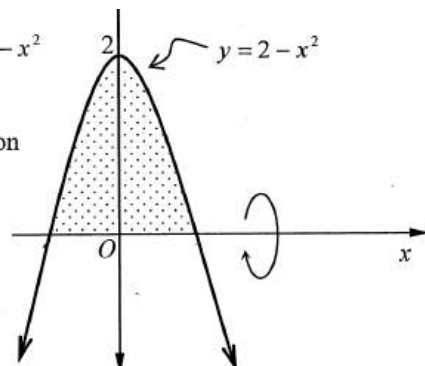
$$\begin{aligned} &= \pi \left[9y - 3y^2 + \frac{y^3}{3} + \frac{4}{y} \right]_1^2 \\ &= \pi \left[\frac{32}{3} - \frac{31}{3} \right] \\ &= \frac{\pi}{3} \text{ u}^3 \text{ (approximately 1.05 to 2 d.p.)} \end{aligned}$$

Q12

CSSA

In the diagram the region bounded by the curve $y = 2 - x^2$ and the x -axis is shaded.

Find the volume of the solid that is formed if this region is rotated about the x -axis.



Sample answer

$y = 2 - x^2$, cuts the x -axis when $y = 0$.

$$0 = 2 - x^2$$

$$x = \pm\sqrt{2}$$

$$V = 2\pi \int_0^{\sqrt{2}} (4 - 4x^2 + x^4) dx = 2\pi \left[4x - \frac{4x^3}{3} + \frac{x^5}{5} \right]_0^{\sqrt{2}} = 2\pi \left[4\sqrt{2} - \frac{8\sqrt{2}}{3} + \frac{4\sqrt{2}}{5} \right]$$

$$= \frac{64\pi\sqrt{2}}{15} \approx 18.96 \text{ (2 d.p.)}$$

Q13

GIRAWEN

The area under the curve $y = \frac{5}{\sqrt{x}}$, for $1 \leq x \leq e^3$, is rotated about the x axis. What is the exact volume of the solid of revolution?

(A) 15π units³

(B) 25π units³

(C) 28π units³

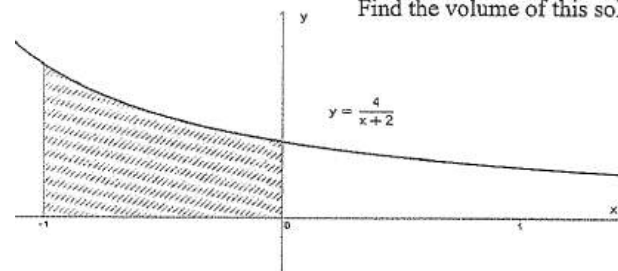
(D) 75π units³

Q14

GIRAWEN

The region bounded by the curve $y = \frac{4}{x+2}$, the line $x = -1$ and the axes is shown below.

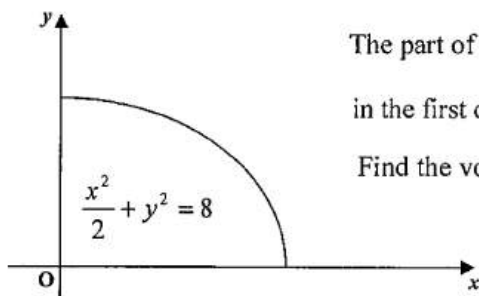
The region is rotated about the x -axis to form a solid. Find the volume of this solid.



$$\begin{aligned} \textcircled{a} V &= \pi \int_{-1}^0 y^2 dx \\ &= \pi \int_{-1}^0 \frac{16}{(x+2)^2} dx \\ &= 16\pi \int_{-1}^0 (x+2)^{-2} dx \\ &= 16\pi \left[-(x+2)^{-1} \right]_{-1}^0 \\ &= 16\pi \left[\frac{-1}{x+2} \right]_{-1}^0 \\ &= 16\pi \left[\left(-\frac{1}{2} \right) - (-1) \right] \\ &= 8\pi \text{ units}^3 \end{aligned}$$

Q15

GOSFORD



The part of the curve $\frac{x^2}{2} + y^2 = 8$ that lies in the first quadrant is rotated about the x -axis.
Find the volume of the solid of revolution.

$$\frac{x^2}{2} + y^2 = 8$$

$$y^2 = 8 - \frac{x^2}{2}$$

$$V = \pi \int y^2 dx$$

$$= \pi \int_0^4 \left(8 - \frac{x^2}{2}\right) dx$$

$$= \pi \left[8x - \frac{x^3}{6} \right]_0^4$$

$$= \pi \left[32 - \frac{64}{6} \right]$$

$$= \frac{64\pi}{3} \text{ cubic units}$$

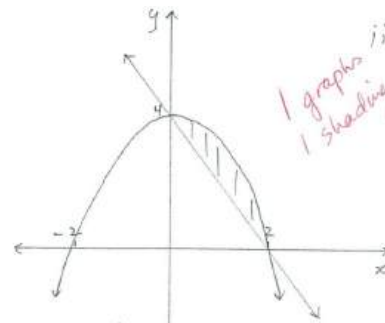
Q16

THG

The area enclosed between the curve $y = 4 - x^2$ and the line $y = 4 - 2x$ is rotated about the x -axis.

- Sketch the region between the two graphs.
- Find the volume of the solid generated between these two graphs, leaving your answer in terms of π .

d) i)



ii)

$$V = \pi \int_{-2}^2 (4 - x^2)^2 dx - \pi \int_{-2}^2 (4 - 2x)^2 dx$$

$$= \pi \int_{-2}^2 (16 - 8x^2 + x^4) - (16 - 16x + 4x^2) dx$$

$$= \pi \int_{-2}^2 (16 - 8x^2 + x^4 - 16 + 16x - 4x^2) dx$$

$$= \pi \int_{-2}^2 (-12x^2 + 16x + x^4) dx$$

$$= \pi \left[-\frac{12x^3}{3} + \frac{16x^2}{2} + \frac{x^5}{5} \right]_{-2}^2$$

$$= \pi \left(\frac{(2)^5}{5} - 4(2)^3 + 8(2)^2 \right)$$

$$= \frac{32\pi}{5} \text{ units}^3$$